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1 Decay Scheme

²¹²Pb decays by β^- emission to the 1^- ground state and two excited states (238.6 keV, $J^\pi = 0^-$ and 415.3 keV, $J^\pi = 1^-$) of ²¹²Bi.

Le ²¹²Pb décroît par émission β^- vers l'état fondamental 1^- et deux états excités ($J^\pi = 0^-$ à 238,6 keV et $J^\pi = 1^-$ à 415,3 keV) du ²¹²Bi.

2 Nuclear Data

$T_{1/2}(\text{²¹²Pb})$: 10,630 (7) h
 $T_{1/2}(\text{²¹²Bi})$: 60,54 (6) min
 $Q^-(\text{²¹²Pb})$: 569,0 (18) keV

2.1 β^- Transitions

	Energy (keV)	Probability (%)	Nature	log <i>ft</i>
$\beta_{0,3}^-$	153,7 (18)	4,92 (8)	1st Forbidden	5,45
$\beta_{0,2}^-$	330,4 (18)	80,8 (14)	1st Forbidden	5,28
$\beta_{0,0}^-$	569,0 (18)	14,4 (14)	1st Forbidden	6,80

2.2 Gamma Transitions and Internal Conversion Coefficients

	Energy (keV)	P _{$\gamma+ce$} (%)	Multipolarity	α_K	α_L	α_M	α_T
$\gamma_{1,0}(\text{Bi})$	115,183 (5)	4,57 (8)	M1	5,53 (8)	0,972 (14)	0,229 (4)	6,8 (1)
$\gamma_{3,2}(\text{Bi})$	176,640 (9)	0,162 (7)	M1	1,646 (23)	0,287 (4)	0,0675 (10)	2,02 (3)
$\gamma_{2,0}(\text{Bi})$	238,632 (2)	81 (1)	M1	0,71 (1)	0,1232 (18)	0,0290 (4)	0,872 (13)
$\gamma_{3,1}(\text{Bi})$	300,089 (10)	4,74 (5)	M1	0,378 (6)	0,0653 (10)	0,01534 (22)	0,464 (7)
$\gamma_{3,0}(\text{Bi})$	415,272 (9)	0,0155 (36)	(M1)	0,1571 (22)	0,0269 (4)	0,00632 (9)	0,192 (3)

3 Atomic Data

3.1 Bi

ω_K	:	0,964	(4)
$\bar{\omega}_L$	:	0,391	(16)
n_{KL}	:	0,809	(5)

3.1.1 X Radiations

		Energy (keV)	Relative probability
X _K			
	K α_2	74,8157	59,77
	K α_1	77,1088	100
	K β_3	86,835	34,25391
	K β_1	87,344	
	K β_5''	87,862	
	K β_2	89,732	10,48169
	K β_4	90,074	
	KO _{2,3}	90,421	
X _L			
	L ℓ	9,4207	
	L α	10,7308 - 10,8387	
	L η	11,7127	
	L β	12,4814 - 13,8066	
	L γ	14,7735 - 15,7084	

3.1.2 Auger Electrons

		Energy (keV)	Relative probability
Auger K			
	KLL	57,491 - 63,419	100
	KLX	70,025 - 77,105	56
	KXY	82,53 - 90,52	7,84
Auger L		5,4204 - 16,3366	

4 Electron Emissions

		Energy (keV)	Electrons (per 100 disint.)	
e _{AL}	(Bi)	5,4204 - 16,3366	21,20 (26)	
e _{AK}	(Bi)		}	1,27 (15)
	KLL	57,491 - 63,419		
	KLX	70,025 - 77,105		
	KXY	82,53 - 90,52		
ec _{1,0} T	(Bi)	24,657 - 115,159	3,99 (7)	
ec _{1,0} K	(Bi)	24,657 (5)	3,24 (6)	
ec _{3,2} T	(Bi)	86,11 - 176,62	0,1085 (45)	
ec _{1,0} L	(Bi)	98,796 - 101,764	0,57 (1)	
ec _{1,0} M	(Bi)	111,184 - 112,603	0,1342 (27)	
ec _{2,0} T	(Bi)	148,106 - 238,608	37,7 (7)	
ec _{2,0} K	(Bi)	148,106 (2)	30,7 (5)	
ec _{3,1} T	(Bi)	209,563 - 300,065	1,502 (27)	
ec _{3,1} K	(Bi)	209,563 (12)	1,224 (23)	
ec _{2,0} L	(Bi)	222,245 - 225,213	5,33 (9)	
ec _{2,0} M	(Bi)	234,633 - 236,052	1,255 (21)	
ec _{2,0} N	(Bi)	237,694 - 238,475	0,321 (6)	
ec _{3,1} L	(Bi)	283,702 - 286,670	0,2114 (38)	
$\beta_{0,3}^-$	max:	153,7 (18)	}	4,92 (8)
	avg:	40,6 (5)		
$\beta_{0,2}^-$	max:	330,4 (18)	}	80,8 (14)
	avg:	92,7 (6)		
$\beta_{0,0}^-$	max:	569,0 (18)	}	14,4 (14)
	avg:	170,6 (6)		

5 Photon Emissions

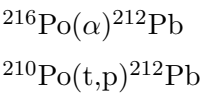
5.1 X-Ray Emissions

		Energy (keV)	Photons (per 100 disint.)		
XL	(Bi)	9,4207 - 15,7084	13,46 (24)		
XKα ₂	(Bi)	74,8157	9,94 (17)	}	Kα
XKα ₁	(Bi)	77,1088	16,63 (27)		
XKβ ₃	(Bi)	86,835	5,69 (13)	}	K'β ₁
XKβ ₁	(Bi)	87,344			
XKβ ₅ ''	(Bi)	87,862			
XKβ ₂	(Bi)	89,732	1,74 (5)	}	K'β ₂
XKβ ₄	(Bi)	90,074			
XKO _{2,3}	(Bi)	90,421			

5.2 Gamma Emissions

	Energy (keV)	Photons (per 100 disint.)
$\gamma_{1,0}(\text{Bi})$	115,183 (5)	0,5862 (61)
$\gamma_{3,2}(\text{Bi})$	176,640 (9)	0,0537 (21)
$\gamma_{2,0}(\text{Bi})$	238,632 (2)	43,26 (42)
$\gamma_{3,1}(\text{Bi})$	300,089 (10)	3,238 (31)
$\gamma_{3,0}(\text{Bi})$	415,272 (9)	0,013 (3)

6 Main Production Modes



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